DisputeDesk AI — Phase 5

Dispute Prevention & Resolution Intelligence

Baseline 90cbe40 (master) **Date** 4 November 2026 **Status** implemented, not committed

Summary. Merchants and admin now have an **Insights** view: fixed rules turn the account's synthetic demo disputes into counts, patterns and six kinds of suggestion, each with what was detected, the disputes and record fields behind it, why it matters, a practical action and its limits. Percentages appear only when at least 10 disputes are behind a share; a pattern needs the same gap in at least 3 disputes; a monthly view needs at least 6 dated disputes across 3 months. Gemini may only reword the findings, and every paragraph is checked for invented numbers, percentages, causes, predictions, trends, dates and names. Buyers get 403. No live PayPal or Gemini calls were made.

179/179	63/63	2 × 48/48	clean	pass
server tests (was 159)	client tests (was 56)	browser checks, 4173 and 5173	both type checks	production build

1. Skills and plugins used

Discovered by listing the installed plugins, user skills and hooks. Nothing was installed or connected.

Skill	Used for
`dataviz`	Form choice (bar rows and a table rather than pie charts), counts as text on every bar, a table view, no dual axes.
`api-and-interface-design`	Contract-first shared types, the existing error shape, additive endpoints, validation at the boundary.
`security-and-hardening`	Per-account scope from the server's session, LLM output treated as untrusted, no free record text in the prompt.
`test-driven-development`	Engine and migration tests written first and seen failing; mutation check on tenant isolation.
`frontend-ui-engineering`, `ponytail`	Accessible responsive UI with existing tokens and components; no chart library or new dependency.

The `ecc:chrome-devtools` plugin did not connect (timeout), so the existing headless-Chrome script was used for browser checks.

2. Files changed and architecture

19 files, 2041 insertions, 42 deletions. 3 new files.

File	Change

`server/src/intelligence.ts` (new)	The rules engine: patterns, recommendations, trend, notes and thresholds, built on `analyzeDispute` and `findContradictions`. Also the Gemini request, the insights wording checks and the standard summary.
`server/src/intelligence.test.ts` (new)	13 tests on known synthetic fixtures.
`client/src/Intelligence.tsx` (new)	Insights view: summary tiles, data notes, suggestion cards, patterns, evidence table, status and reasons, monthly view, summary button; loading, error and empty states.
`server/src/app.ts`	`GET /api/intelligence` (merchant, admin) and `POST /api/intelligence/explain` (merchant).
`server/src/db.ts`	Migration 3: audit events `intelligence_viewed` and `intelligence_explanation_generated`. Each released event list is frozen; the audit-log rebuild is a helper used only by migration 3 (migration 2's SQL hash is unchanged and now pinned in a test).
`shared/types.ts`, `shared/disputes.ts`	Insights types; `MIN_FOR_PERCENT` and `formatShare`, the single rule for when a percentage may be shown.
`demo-disputes.json`, `demo-accounts.json`	Demo Merchant C (scrypt hash only) and 14 synthetic disputes, DEMO-2001 to DEMO-2014.
`server/src/buyer.ts`	Exports its promise, legal and blame checks for reuse; no behaviour change.
`client/src/App.tsx`, `sandbox.ts`, `styles.css`	Disputes / Insights switch for merchants and admin; checked fetch helpers; bar, card and table styles.
Tests and docs	`app.test.ts`, `db.test.ts`, `store.test.ts`, `ui.test.ts`; README and CLAUDE.md.

Flow. Session → account (server) → `loadDisputes` filtered by `canAccess` → `buildIntelligence` (demo records only) → counts and dispute IDs as JSON → the browser formats shares with `formatShare`. The server never sends a percentage.

3. Intelligence rules and their evidence sources

Pattern	Detected when	Record fields
Delivery proof gap	Item-not-received dispute where proof of delivery is not Available (engine status)	`reason`, `evidence.proof_of_delivery`
Tracking gap	Order marked shipped or delivered, tracking not Available	`order.shipping_status`, `order.tracking_number`, `evidence.shipping_tracking`
Communication gap	Customer communication not Available	`evidence.customer_communication`
Order record gap	Order confirmation not Available	`evidence.order_confirmation`
Credit disputes	Reason is Credit not processed	`reason`

Conflicting records	The evidence engine finds a contradiction	`evidence`, `order`
Waiting for seller	Status WAITING_FOR_SELLER_RESPONSE	`status`
Open over 30 days	Unresolved and opened more than 30 days before today; undated disputes are left out	`create_time`, `status`

Suggestion	Shown when
Improve delivery confirmation records	Delivery proof gap in at least 3 disputes
Keep a tracking record for every shipment	Tracking gap in at least 3
Keep clearer customer communication records	Communication gap in at least 3
Keep transaction and refund records organized	Order record gap in at least 3, or at least 2 credit disputes
Resolve conflicting records	Conflicts in at least 2 disputes
Review unresolved disputes	At least 1 waiting for the seller, or at least 1 open over 30 days

Confidence is "Moderate" only with at least 10 disputes and at least 3 supporting ones; otherwise "Limited data". It is never "high". Every number in rule-written text is checked by a test to come from the data or a published threshold. All rule text passes the same checks as AI wording: no "prevent", "reduce", "cause", "trend", "win" or promise words. Suggestions say they are prompts to check records, not predictions of any outcome.

4. Dashboard screenshots

DD DisputeDesk

Demo Merchant C Merchant Sign out

DEMO DATA — NOT A REAL PAYPAL DISPUTE

Insights use only the synthetic demo disputes on this account. PayPal sandbox disputes are not included, and nothing here is connected to PayPal.

Disputes | Insights

Disputes checked	Unresolved	Recurring patterns	Suggestions
14	10 of 14 (71%)	6	6

About this data

Your demo disputes

- Based on 14 synthetic demo disputes on this account. PayPal sandbox disputes are not included.
- A gap counts as a pattern only when it appears in at least 3 disputes.
- 1 dispute has no opened date, so it is left out of the monthly view and the age check.
- No outcome or resolved date is on record, so outcome rates and time to resolution are not shown.
- Evidence comes from the flags on each record. DisputeDesk does not inspect documents, and a gap in the records does not show what happened.
- Ages are measured to Nov 4, 2026.

Suggestions to check

Improve delivery confirmation records

Moderate confidence

WHAT WAS DETECTED

Proof of delivery is missing or unverified in 6 of 7 item-not-received disputes.

SUPPORTING RECORDS

`DEMO-2003, DEMO-2005, DEMO-2007, DEMO-2009, DEMO-2011, DEMO-2013`
`evidence.proof_of_delivery, reason`

WHY IT MATTERS

For item-not-received disputes, a carrier delivery confirmation is key evidence. Without one on file, a response has no carrier record of the delivery to point to.

PRACTICAL ACTION

Save the carrier's delivery confirmation with each order, so it is on file if a buyer reports an item not received.

LIMITATION

Based only on the evidence flags on these synthetic demo records. DisputeDesk does not inspect documents, and a gap in the records does not show what happened. A missing flag means the record was not supplied, not that an order was not delivered.

Keep a tracking record for every shipment

Moderate confidence

WHAT WAS DETECTED

3 of 11 orders marked shipped or delivered have no usable tracking record.

SUPPORTING RECORDS

`DEMO-2003, DEMO-2007, DEMO-2011`
`evidence.shipping_tracking, order.shipping_status, order.tracking_number`

WHY IT MATTERS

A carrier tracking record can show that an order was sent and how it moved.

PRACTICAL ACTION

Record the carrier tracking number when an order ships, and check that it is attached before marking the order shipped.

LIMITATION

Based only on the evidence flags on these synthetic demo records. DisputeDesk does not inspect documents, and a gap in the records does not show what happened.

Keep clearer customer communication records

Moderate confidence

WHAT WAS DETECTED

Customer communication is missing or unverified in 5 of 14 disputes.

SUPPORTING RECORDS

`DEMO-2002, DEMO-2006, DEMO-2008, DEMO-2009, DEMO-2012`
`evidence.customer_communication`

WHY IT MATTERS

Messages with the buyer can show what was offered, reported or agreed.

PRACTICAL ACTION

Keep order-related messages with the order, and answer buyer questions in writing so there is a record.

LIMITATION

Based only on the evidence flags on these synthetic demo records. DisputeDesk does not inspect documents, and a gap in the records does not show what happened.

Keep transaction and refund records organized

Limited data

WHAT WAS DETECTED

2 of 14 disputes are about a credit or refund not processed.

SUPPORTING RECORDS

`DEMO-2004, DEMO-2010`
`reason`

WHY IT MATTERS

Order confirmations show what the buyer agreed to pay for, and refund records show whether money was returned.

PRACTICAL ACTION

Keep order confirmations, refund confirmations and return records together for each order, so they can be found quickly.

LIMITATION

Based only on the evidence flags on these synthetic demo records. DisputeDesk does not inspect documents, and a gap in the records does not show what happened. This rests on few records, so treat it as a prompt to check rather than a pattern.

Resolve conflicting records

Moderate confidence

WHAT WAS DETECTED

3 of 14 disputes have records that contradict each other.

SUPPORTING RECORDS

`DEMO-2005, DEMO-2007, DEMO-2009`
`evidence, order`

WHY IT MATTERS

A response can't rely on records that contradict each other until they are checked.

PRACTICAL ACTION

Check each order's shipping status and evidence flags against your order system and the carrier, and correct whichever record is wrong.

LIMITATION

Based only on the evidence flags on these synthetic demo records. DisputeDesk does not inspect documents, and a gap in the records does not show what happened.

Review unresolved disputes

Moderate confidence

WHAT WAS DETECTED

3 disputes are waiting for the seller's response. 7 of 9 unresolved disputes with an opened date have been open for more than 30 days.

SUPPORTING RECORDS

`DEMO-2005, DEMO-2006, DEMO-2007, DEMO-2008, DEMO-2009, DEMO-2010, DEMO-2011, DEMO-2012, DEMO-2014`
`create_time, status`

WHY IT MATTERS

A dispute waiting for the seller usually has a response deadline set by the payment provider, and DisputeDesk does not have that deadline on record.

PRACTICAL ACTION

Open each dispute listed here, check its deadline in PayPal, and prepare a response from your records.

LIMITATION

Based only on the evidence flags on these synthetic demo records. DisputeDesk does not inspect documents, and a gap in the records does not show what happened. Age is measured from the opened date only; DisputeDesk has no record of later activity.

Suggestions are prompts to check your own records. They do not predict or change any dispute's outcome.

Patterns found

Proof of delivery missing or unverified in item-not-received disputes — Recurring
6 of 7
DEMO-2003, DEMO-2005, DEMO-2007, DEMO-2009, DEMO-2011, DEMO-2013

Shipped or delivered orders without usable tracking — Recurring
3 of 11 (27%)
DEMO-2003, DEMO-2007, DEMO-2011

Customer communication missing or unverified — Recurring
5 of 14 (36%)
DEMO-2002, DEMO-2006, DEMO-2008, DEMO-2009, DEMO-2012

Order confirmation missing or unverified — Below pattern minimum
2 of 14 (14%)
DEMO-2006, DEMO-2010

Disputes about a credit or refund not processed — Below pattern minimum
2 of 14 (14%)
DEMO-2004, DEMO-2010

Records that contradict each other — Recurring
3 of 14 (21%)
DEMO-2005, DEMO-2007, DEMO-2009

Waiting for the seller's response — Recurring
3 of 14 (21%)
DEMO-2011, DEMO-2012, DEMO-2014

Unresolved for more than 30 days since opened — Recurring
7 of 9
DEMO-2005, DEMO-2006, DEMO-2007, DEMO-2008, DEMO-2009, DEMO-2010, DEMO-2011

Evidence completeness

Statuses from the evidence engine, across 14 disputes.

Evidence	Available	Unverified	Missing
Order confirmation	12 of 14 (86%)	0 of 14 (0%)	2 of 14 (14%) DEMO-2006, DEMO-2010
Shipping tracking	8 of 14 (57%)	4 of 14 (29%) DEMO-2007, DEMO-2008, DEMO-2010, DEMO-2014	2 of 14 (14%) DEMO-2003, DEMO-2011
Proof of delivery	5 of 14 (36%)	4 of 14 (29%) DEMO-2007, DEMO-2008, DEMO-2010, DEMO-2014	5 of 14 (36%) DEMO-2003, DEMO-2005, DEMO-2009, DEMO-2011, DEMO-2013
Customer communication	9 of 14 (64%)	1 of 14 (7%) DEMO-2008	4 of 14 (29%) DEMO-2002, DEMO-2006, DEMO-2009, DEMO-2012

Status

Resolved 4 · outcome not on record

Open — 2 of 14 (14%)
Awaiting seller — 3 of 14 (21%)
Awaiting buyer — 2 of 14 (14%)
Under review — 3 of 14 (21%)
Resolved — 4 of 14 (29%)

Reasons

Item not received — 7 of 14 (50%)
Not as described — 3 of 14 (21%)
Credit not processed — 2 of 14 (14%)
Duplicate transaction — 1 of 14 (7%)
Unauthorized transaction — 1 of 14 (7%)

By month opened

2026-05
2 disputes · 7 of 8 evidence items available

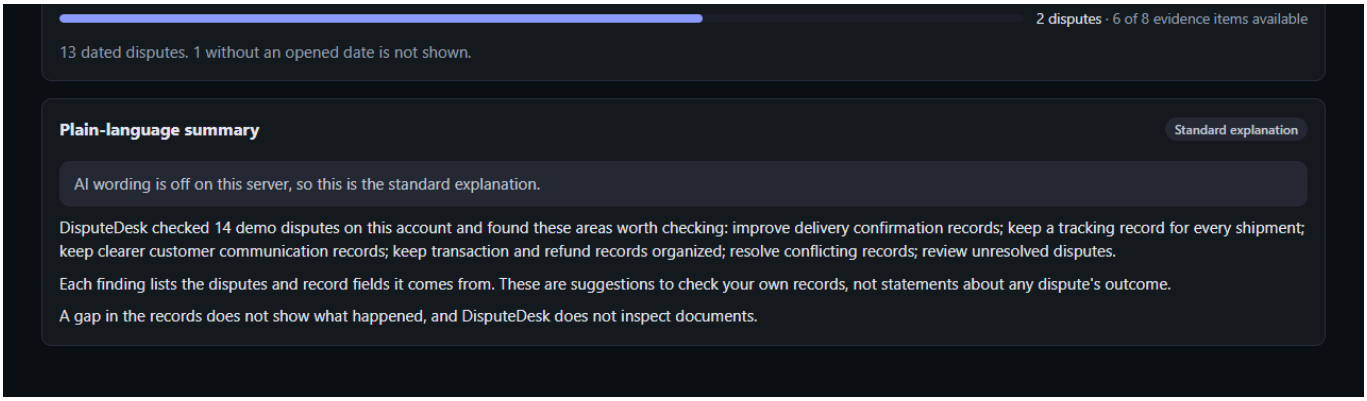
2026-06
2 disputes · 6 of 8 evidence items available

2026-07
3 disputes · 7 of 12 evidence items available

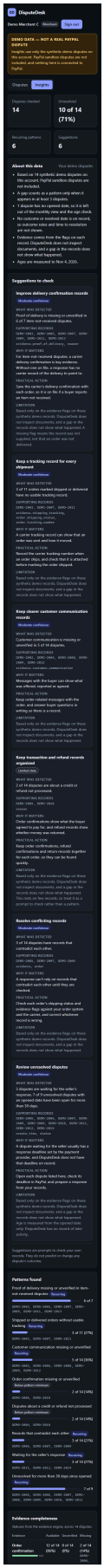
2026-08
2 disputes · 3 of 8 evidence items available

2026-09
2 disputes · 3 of 8 evidence items available

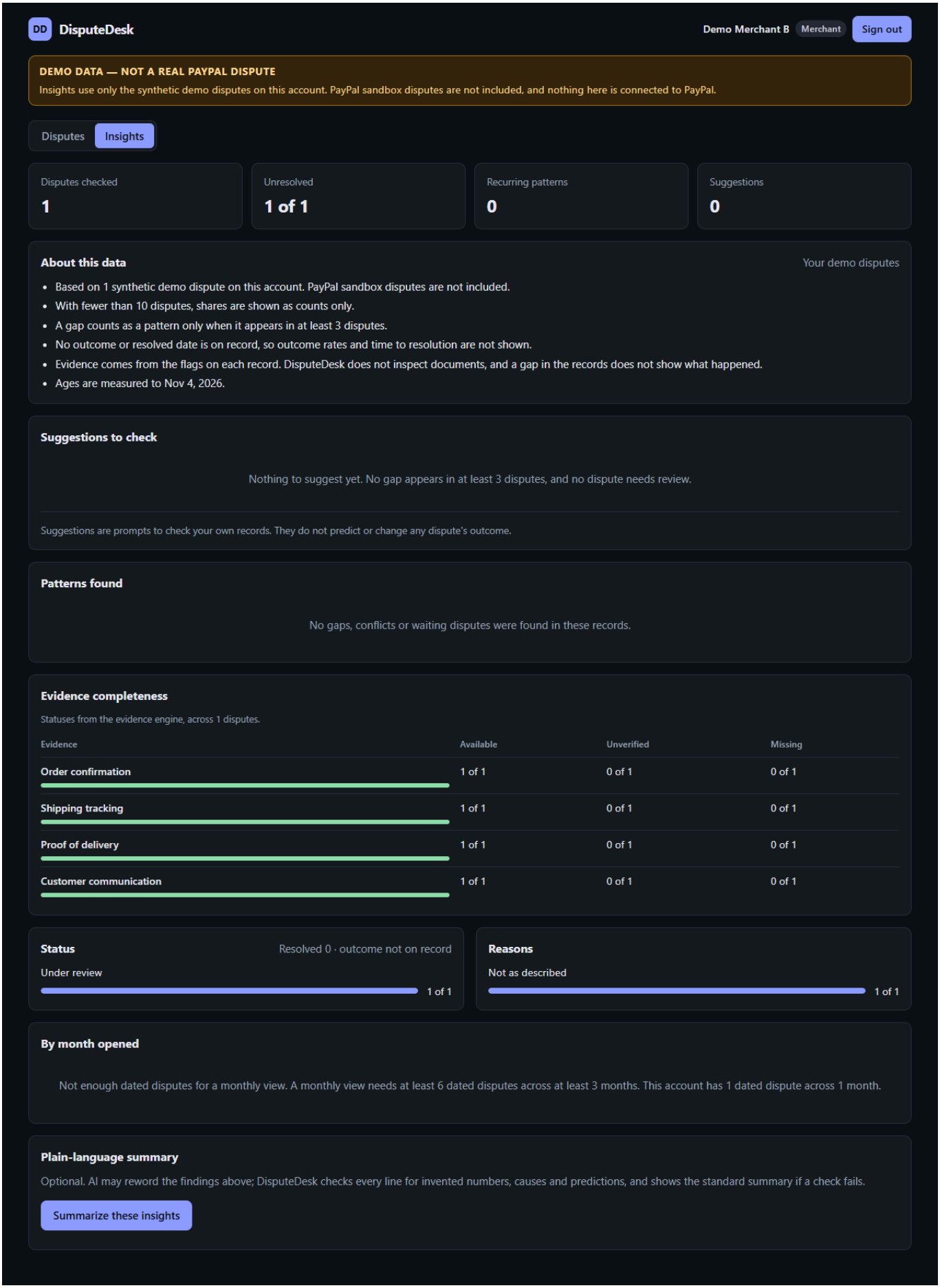
2026-10



Merchant C, 1280 px (14 synthetic disputes). Six suggestion cards with supporting dispute IDs and record fields. "6 of 7" has no percentage because only 7 disputes are behind it; "3 of 11 (27%)" does. The summary is the labelled standard explanation, because AI is off on this server.



Merchant C, 375 px. Cards and bars stack; the evidence table fits. No horizontal scroll at 375 or 320 px.



Merchant B, 1 dispute. "Nothing to suggest yet", counts only, no percentages, and "Not enough dated disputes for a monthly view".

5. Browser evidence

Headless Chrome, signed in as each demo role on the production build (port 4173) and the dev server (port 5173). Both give the same 48 results. The first run on 4173 recorded no response for the cross-site sign-out probe (a timing miss in the script, 47/48); the rerun passed 48/48, as did the run after the last code change.

- ✓ page headers: X-Frame-Options, nosniff, Referrer-Policy
- ✓ signed out: /api/auth/me is 401
- ✓ signed out: /api/health is 401
- ✓ cookie flags: HttpOnly=true Secure=true SameSite=Strict Path=/; no Domain attribute (host-only); browser storage host localhost
- ✓ document.cookie is empty (HttpOnly)
- ✓ API headers through the proxy
- ✓ Merchant A sees DEMO-1001 and DEMO-1002 only
- ✓ Merchant A: DEMO-1003 is 404
- ✓ Merchant A: health 200
- ✓ Merchant A: analysis and validation example render
- ✓ page fetch to another origin: blocked
- ✓ cross-site form POST to sign-out is refused: 403
- ✓ and the session is unaffected
- ✓ sign-out removes the cookie from the browser
- ✓ replaying the signed-out cookie: 401
- ✓ Buyer B sees DEMO-1002 and DEMO-1003 only
- ✓ Buyer B: no merchant order, tracking, message, evidence or draft on the page
- ✓ Buyer B: one current stage, marked with aria-current=step
- ✓ Buyer B: source labels shown
- ✓ Buyer B: explaining DEMO-1001 is 404
- ✓ Buyer B: explanation shown (standard)
- ✓ Buyer B at 1280px: no horizontal scroll
- ✓ Buyer B at 375px: no horizontal scroll
- ✓ Buyer B at 320px: no horizontal scroll
- ✓ Buyer B: health and merchant list 403
- ✓ Buyer B: insights and insights summary 403
- ✓ Buyer B: no Insights view
- ✓ Merchant C sees its 14 demo disputes
- ✓ Merchant C insights: demo label and sandbox exclusion shown
- ✓ Merchant C insights: six suggestions, each a card
- ✓ every suggestion shows detected, records, why, action and limitation
- ✓ a share of 7 is shown as a count, not a percentage
- ✓ no percentage with fewer than 10 disputes behind it
- ✓ no order item, order ID or buyer message in insights
- ✓ every bar states its count in text
- ✓ view switch buttons expose aria-pressed
- ✓ Merchant C: summary shown (standard)
- ✓ Merchant C insights at 1280px: no horizontal scroll
- ✓ Merchant C insights at 375px: no horizontal scroll
- ✓ Merchant C insights at 320px: no horizontal scroll
- ✓ Merchant B (1 dispute): nothing suggested, no monthly view, no percentages
- ✓ Merchant B's insights response holds no other merchant's dispute
- ✓ Admin: no Generate AI draft button
- ✓ Admin: AI route 403
- ✓ Admin: DELETE accounts 404
- ✓ Admin insights: all merchants, read-only, no AI summary (403)
- ✓ CSP violations since the last page load: none
- ✓ console errors other than expected 4xx: none

6. Privacy and security results

- **Tenant isolation:** each merchant's insights contain only their own disputes (tested for A, B and C). Query strings and request bodies naming another account or scope change nothing. Removing the access filter made 5 tests fail; it was restored.
- **Buyers:** 403 on both routes, with no dispute ID in the refusal; no Insights button; `Buyer.tsx` never names the endpoint.
- **Admin:** reads all demo merchants, labelled read-only; the summary route is 403.
- **No record free text:** insights and the Gemini prompt contain no buyer message, order item or order ID. Dispute IDs that read like instructions are withheld from the prompt.
- **PayPal sandbox:** never included; the insights routes make no PayPal call.
- **AI:** wording is used only if every paragraph passes; the failed text is never returned. Off, erroring and failing cases fall back to the labelled standard summary. One request per merchant at a time (409).
- **Unchanged:** security headers, strict production CSP (no violations), Origin and Sec-Fetch-Site checks, session cookie flags, parameterized SQL, append-only audit log. Migration 3 kept all earlier audit rows in the live database (now schema 1, 2, 3).

- **Secrets:** no `.env` change. The new demo password is only in README.md (where the demo passwords live) and the scratchpad file; it is not in this report, the screenshots or chat.

7. Exact test counts

Suite	Result
Server (`npm test`)	179 pass, 0 fail: 13 new engine tests, 6 new API tests plus 2 role-matrix rows, 1 new migration test, 1 extended migration-hash test; 7 older tests updated for the larger demo set
Client (`npm test`)	63 pass, 0 fail: 7 new insights UI tests; 3 older tests updated (demo-set size, endpoint list)
Type checks	server and client clean
Production build	pass (264 kB JS, 13.6 kB CSS before gzip)
Browser	48/48 on 4173 and 48/48 on 5173

8. Remaining limitations

- **Synthetic data only.** 14 records for Merchant C show how the rules behave, not real business insight.
- **No outcome data.** No outcome or resolved date is on record, so there are no outcome rates or time to resolution.
- **AI number check is set-based.** It confirms each number appears in the findings, not that it is paired with the right finding.
- AI path tested with a mocked Gemini only; AI is off on this server (`GEMINI_MODEL` invalid).
- Evidence is four yes/no flags per dispute; documents are not inspected.
- The monthly view groups by the opened date in UTC.
- `npm audit` and the optional security-reviewer agent still need your approval.

9. Git status and commit details

Branch `master` at `90cbe40`. 16 tracked files modified and 3 new files (`server/src/intelligence.ts` , `server/src/intelligence.test.ts` , `client/src/Intelligence.tsx`), 2041 insertions and 42 deletions in total. `reports/` stays untracked. Nothing is staged, committed or pushed.